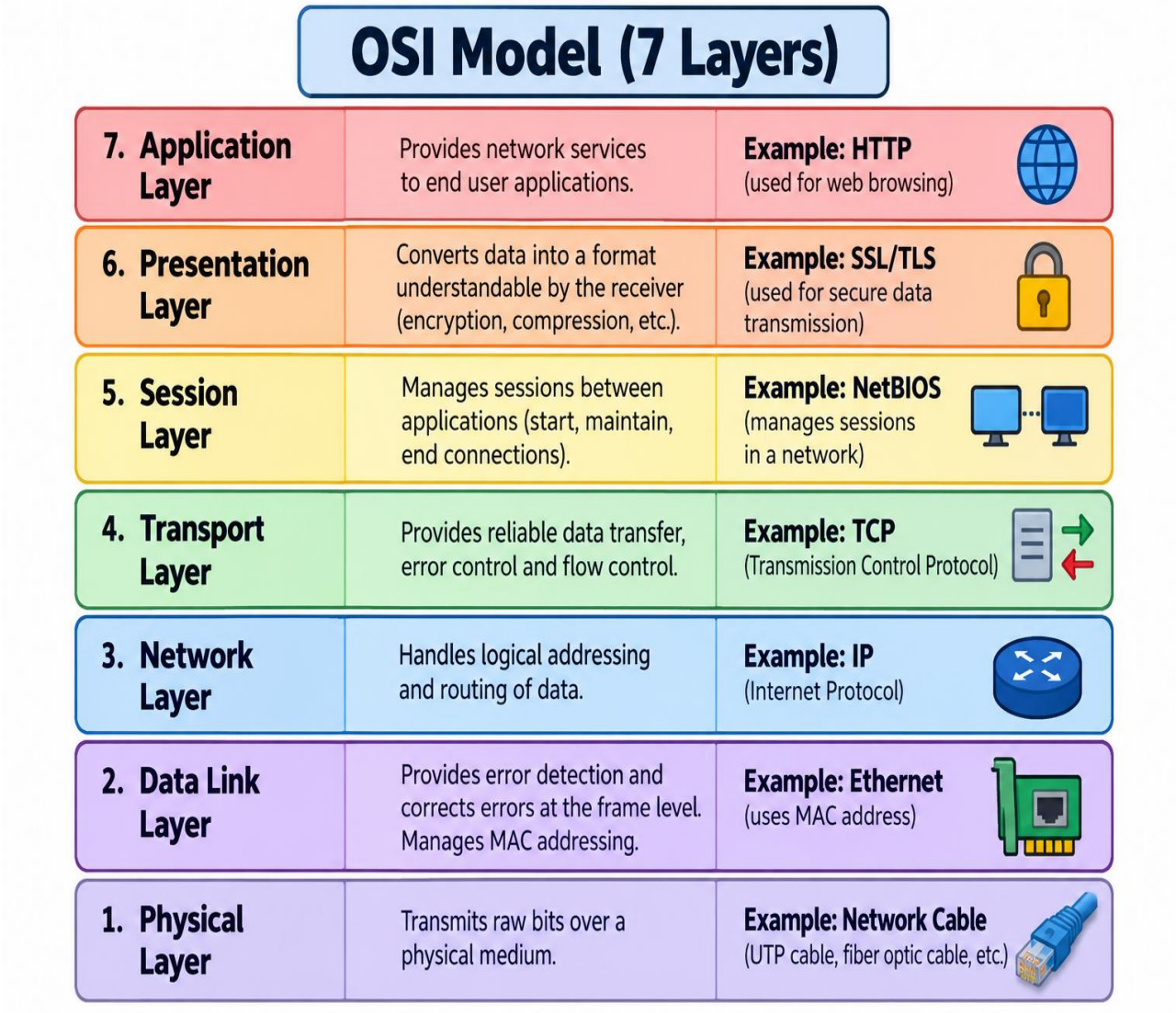
13.introduction to networking and the OSI model :

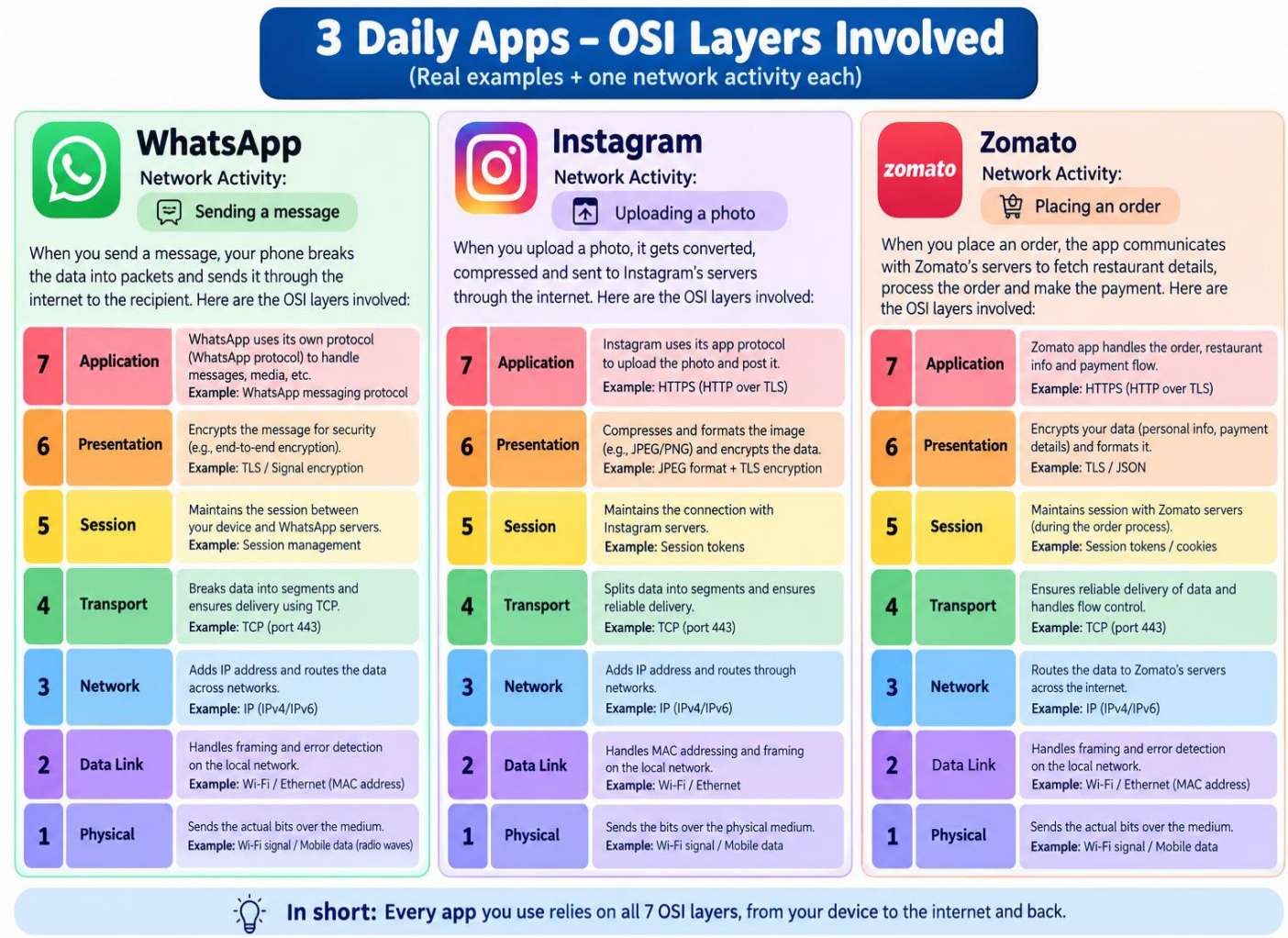
1. **Draw an OSI Model diagram showing all 7 layers, and for each layer, write one real-world example (such as HTTP for Application, TCP for Transport, Ethernet for Data Link, cables for Physical, etc).**

ANS:



2. **List three apps you use daily (for example: Instagram, Zomato, or WhatsApp) and for each, pick one network activity (like sending a message or placing an order) and explain which OSI layers are involved in that activity.**

ANS



3. **Install and open Wireshark on your computer, start a capture, then visit any website (for example, www.example.com) in your browser. Stop the capture and save the PCAP file.**

ANS:

Steps in Wireshark

1. Install Wireshark
    - Download and install Wireshark from the official website: Wireshark official download
    - During installation, allow Npcap to be installed when prompted. It is required for capturing network traffic.
2. Open Wireshark
    - Start Wireshark.
    - You will see network interfaces such as Wi-Fi or Ethernet.
3. Start capture
    - If you are connected through Wi-Fi, double-click Wi-Fi.
    - Packets will start appearing.
4. Visit a website
    - While the capture is running, open your browser.
    - Visit: www.example.com
    - Wait for the page to load completely.
5. Stop capture
    - Go back to Wireshark.
    - Click the red square (Stop) button at the top.
6. Save the PCAP file
    - Click File → Save As.
    - Choose a location such as Desktop.
    - Name it: wireshark_capture.pcapng
    - Click Save.

For your assignment

You should have:

PCAP file: wireshark_capture.pcapng

You can then submit this file to Moodle along with your assignment if required.

4. **In Wireshark, find a captured HTTP request packet from your browsing session, take a screenshot of the packet details, and annotate which parts of the packet correspond to the Application, Transport, Network, and Data Link layers.

Hint: Use the packet details pane in Wireshark to expand each protocol section (HTTP, TCP, IP, Ethernet).**

ANS:

1. Find the HTTP packet

In the top **Filter** box, type:

http

Press **Enter**.

Look for a packet showing something like:

GET / HTTP/1.1

or

GET /index.html HTTP/1.1

Select that packet.

Important: If you visited https://..., you may not see HTTP because HTTPS encrypts the HTTP data. For this assignment, use a packet that actually shows **HTTP**.

2. Expand the Packet Details pane

In the middle pane, expand these sections:

- Hypertext Transfer Protocol
- Transmission Control Protocol
- Internet Protocol Version 4
- Ethernet II

3. Annotate the four OSI layers

OSI Layer	Wireshark section	What to point out
Application – Layer 7	**HTTP**	GET, Host, Request URI, HTTP version
Transport – Layer 4	**TCP**	Source Port, Destination Port, Sequence Number
Network – Layer 3	**IPv4**	Source IP, Destination IP
Data Link – Layer 2	**Ethernet II**	Source MAC, Destination MAC, Ethernet Type

4. Take the screenshot

Your screenshot should show:

Packet List → selected HTTP packet → Packet Details with HTTP, TCP, IPv4 and Ethernet II expanded.

Then use Paint/Snipping Tool to add labels such as:

- 🟢 **Application Layer – HTTP**
- 🔵 **Transport Layer – TCP**
- 🟠 **Network Layer – IPv4**
- 🟣 **Data Link Layer – Ethernet II**